

Grace Hui

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EDUCATION

Rensselaer Polytechnic Institute, Troy, NY

Bachelor of Science in Computer Science and Information Technology & Web Science; Minor: Economics

August 2023 – December 2026 (expected)

Dean's List, GPA: 3.72

SKILLS

Programming Languages: Python, Java, C, C++, R, JavaScript, Typescript, SQL

Frameworks and Tools: HTML, CSS, React, API, MySQL, Node.js, MongoDB, Express, Flask, GitHub, Git, Tableau

Cloud Services and Project Management: Azure, AWS, Agile (Scrum), JIRA, Confluence

Relevant Coursework: Data Structures, Introduction to Algorithms, Web Science Systems Development, Machine Learning from Data, Data Mining, Statistical Methods, Database Systems, Data Science, Operating Systems, Introduction to AI, Network Programming

WORK EXPERIENCE

Sumitomo Mitsui Banking Corporation, Jersey City, NJ

Information Technology Intern – Payments and Deposits Technology

June 2026 – August 2026

- Developed a **Python PDF-to-XML parsing solution** that unblocked project delivery when the vendor solution was delayed and became the foundation for the **production implementation**.
- Built **Python and VBA automation tools** to generate sanitized test files from production payment formats (ACH, BAI, COF, Canadian ACH, X9.37), including masking check images for secure testing.
- Performed impact analysis for a Covered Bond branch decommission by developing **Python scripts using pandas** to analyze **IBM Tivoli Workload Scheduler (TWS) job dependencies** and identify downstream system impacts.
- Collaborated with cross-functional teams to implement enhancements across **Oracle** databases and the **Databricks Lakehouse** by documenting business and technical requirements and supporting new data models.
- Traced data flows through legacy payment systems (SQL queries, database views, stored procedures, and source tables) to support modernization and wrote ad hoc **SQL reports** for cash pooling and account validation.

Atlantic Health System, Morristown, NJ

HR Technology Analyst Intern

June 2025 – April 2026

- Conducted data validation and reporting tasks using Oracle OTBI and Microsoft Excel and developed **OTBI data models and dashboards**.
- Reviewed test scripts during **Sprint 4 User Acceptance Testing (UAT)** and identified **critical bugs** involving exposure to sensitive data.
- Authored **Knowledge Articles** while resolving **650+ HR system helpdesk tickets**, delivering support and improving the end-user experience.

Rensselaer Polytechnic Institute, Troy, NY

Undergraduate Mentor

January 2025 – April 2025

- Mentored 40+ students enrolled in **Data Structures** with **debugging code** and explaining concepts to enhance their problem-solving skills.

PROJECTS

Core Flow Movement Studio Booking Platform – Next.js, Resend, Stripe, AWS (Lambda, API Gateway, S3, CloudFront, Cognito, DynamoDB)

- Built and deployed a production serverless booking platform for a Pilates studio, providing an alternative to a third-party service priced at **\$599/month**.
- Developed **end-to-end booking and payment workflows with Amazon Cognito and Stripe**, including memberships, class packs, waitlists, credit tracking, customer accounts, and an admin dashboard.
- Designed and integrated **AWS Lambda APIs with DynamoDB, API Gateway, CloudFront, and S3** to manage application data and deploy.
- Wrote unit tests for booking and payment workflows, covering authentication, credit rules, and input validation.

Locust Infestation Prediction System – Python, Flask, Google Earth Engine, scikit-learn

- Acquired and processed **50,000 satellite/weather data from Google Earth Engine and NASA** sources to create predictions using a Flask backend.
- Trained and evaluated **KNN and Random Forest classifiers**; Random Forest achieved **91% accuracy** versus **74%** for KNN.
- Built interactive **Tableau dashboards** to visualize model performance, prediction errors, feature importance, and environmental patterns.

Craft Corner – React, Typescript, Gemini API, Google Photos API, Pinterest API

- Built an app to organize a digital yarn inventory and recommend projects by matching saved craft ideas from Pinterest against the user's yarn stash.
- Integrated Gemini to extract yarn attributes from uploaded photos and create structured inventory records, alongside Google OAuth and the Google Photos API for uploading craft images to authorized albums.

LEADERSHIP

RPI Girls Who Code, Troy, NY

Technical Lead (December 2025 – Present) | President (December 2024 – December 2025)

December 2024 – Present

- Increased membership by 65%** through events such as corporate information sessions, community events, and workshops as President.
- Built the club website using HTML and CSS, led exam reviews, and guided development workshops as Technical Lead.

HackRPI, Troy, NY

Director of Finance

January 2024 – December 2024

- Managed a **\$18,000 budget for a 500+ participant hackathon** and coordinated with local vendors to obtain food and apparel sponsorships.

CERTIFICATION

- IBM SkillsBuild Data Analytics
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- Certified in Cybersecurity (CC), ISC2, 2361554

August 2026

July 2025

January 2025